A Multimodal Assistive Navigation System Integrating mmWave Radar and Lightweight Vision for Safer Intersection Crossing

Ching-Min Huang, Shih-Ming Wang, Min-Yen Kuo, Chun-Hsiu Yeh, and Xian-Zhong Lin

(Corresponding author: Chun-Hsiu Yeh)

Department of Information Engineering and Computer Science, Feng Chia University
No. 100, Wenhwa Rd., Xitun Dist., Taichung 40724, Taiwan
Email: chunhyeh@fcu.edu.tw

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Abstract

Safe intersection crossing requires visually impaired pedestrians to interpret signals, detect obstacles, and receive timely feedback. This paper presents GuideSense, a multimodal prototype integrating millimeter-wave radar, lightweight vision, Raspberry P_i edge processing, and smartphone voice and vibration feedback. Radar provides obstacle distance and relative motion, while YOLOv8n processes smartphone images to recognize pedestrian signals. A rule-based module combines both outputs to generate safety cues. YOLOv8n achieved an mAP@0.5 of 98.1% on the validation set, and the radar-based 3D CNN achieved a macro F1-score of 89.8%. Five indoor and five outdoor blindfolded simulation trials showed that all modules operated together under controlled conditions. The results demonstrate the prototype's technical feasibility; supervised studies with visually impaired participants are still required to assess real-world effectiveness and safety.

Keywords: Assistive Navigation; Millimeter-Wave Radar; Sensor Fusion; Visually Impaired Users; YOLOv8n

1 Introduction

Crossing urban intersections is one of the most challenging daily mobility tasks for pedestrians with visual impairments. At complex signalized intersections, visually impaired pedestrians may have difficulty locating the correct crosswalk, aligning themselves with the crossing direction, identifying the onset of the walk interval, maintaining a straight path, and completing the crossing before perpendicular traffic begins to move [1]. Safe intersection crossing therefore requires obstacle awareness, signal interpretation, orientation control, timing judgment, and path stability. Traditional non-visual navigation methods, such as white canes, tactile paving, guide dogs, and auditory cues derived from parallel traffic sounds, can support orientation and local obstacle awareness. These methods provide tactile, auditory, or experience-based cues but may not reliably convey real-time information about pedestrian-signal status, crosswalk alignment, crossing direction, and dynamic obstacles [1, 3]. In modern urban environments, wider roads, turning vehicles, complex signal phases, and surrounding noise further reduce the reliability of environmental sound and passive tactile feedback.

Recent studies have explored smartphone-based navigation systems and computer-vision methods for assisting visually impaired pedestrians [3, 5, 11]. Nevertheless, vision-based systems remain sensitive to lighting conditions, occlusion, motion blur, camera orientation, and adverse weather conditions. A complementary sensing modality can provide distance-aware obstacle perception when visual input is unreliable, showing the potential of millimeter-wave (mmWave) radar [6, 8]. However, existing assistive systems rarely integrate visual signal understanding with radar-based obstacle awareness in a unified real-time framework. This gap leads to the core research question of this study: How can we design a lightweight, real-time multimodal assistive navigation system that effectively integrates vision-based signal recognition and radar-based obstacle perception to support safer intersection crossing for visually impaired pedestrians?

Motivated by the complementary strengths of radar sensing and mobile vision, this paper presents GuideSense, a multimodal assistive navigation system designed to support visually impaired pedestrians during intersection crossing. The main contributions of this study are threefold:

- First, we develop a lightweight multimodal framework that combines visual pedestrian-signal recognition with radar-based obstacle perception and delivers real-time feedback through a mobile interface.
- Second, we replace the initial CNN-LSTM radar model with a lightweight 3D CNN architecture to reduce model complexity while retaining short-term motion information.
- Third, we conduct preliminary controlled walking experiments with blindfolded participants in both indoor and outdoor environments to examine the feasibility and integration of the proposed system.

While the proposed system demonstrates technical feasibility, several limitations must be acknowledged. First, the visual model was trained on a limited dataset of 1,000 annotated pedestrian-signal images, which may not fully cover diverse lighting, weather, camera angles, and real-world intersection conditions. Second, the current prototype evaluation focused on technical feasibility and module interaction through blindfolded simulation with sighted participants, rather than clinical effectiveness or real-world safety for actual visually impaired users. Future supervised studies with visually impaired participants are still required to assess true real-world effectiveness.

The remainder of this paper is structured as follows. Section 2 reviews related work on assistive navigation, smartphone vision, and mmWave radar multimodal sensing. Section 3 details the materials and methods of the GuideSense system, including the perception layer, network and edge processing layer, and application layer. Section 4 presents the experimental results covering YOLOv8n performance, radar classification, and integrated system testing. Finally, Section 5 concludes the paper and discusses future work.

2 Related Work

2.1 Assistive Navigation And Electronic Travel Aids

Prior studies on assistive navigation and electronic travel aids have examined how non-visual cues and sensing devices support mobility for visually impaired pedestrians. These approaches support orientation and local obstacle awareness for visually impaired pedestrians. However, crossing complex signalized intersections remains challenging because visually impaired pedestrians may need to locate the correct crosswalk, align themselves with the crossing direction, identify the walk interval, and maintain a stable walking path [1]. A systematic review of urban navigation systems also indicates that outdoor navigation involves multiple tasks, including environment understanding, route planning, real-time guidance, street crossing, and obstacle avoidance [3]. Collectively, these studies identify local obstacle detection, traffic-signal understanding, orientation guidance, and real-time environmental awareness as requirements for intersection assistance. Practical assistive systems must account for sensing accuracy, portability, sensor orientation, latency, false alarms, and the comprehensibility of feedback during walking [2, 3, 6].

2.2 Smartphone Vision For Pedestrian Signal And Crosswalk Recognition

Smartphone-based computer vision suits assistive navigation because smartphones provide cameras, audio output, vibration motors, and accessibility functions. Prior studies have applied computer vision and deep-learning methods to detect pedestrian traffic-light states for crossing assistance [5]. Other work has further investigated lightweight convolutional neural networks for recognizing both pedestrian traffic lights and zebra crossings, showing that visual perception can support real-time intersection guidance [11]. Nevertheless, vision-based approaches remain sensitive to lighting changes, occlusion, motion blur, adverse weather conditions, and camera misalignment, especially when pedestrian signals occupy only a small region in the image or the phone is not properly oriented toward the signal. In safety-critical crossing scenarios, relying only on visual recognition may therefore lead to unstable or delayed feedback. This limitation motivates the use of complementary sensing to provide obstacle-related information beyond camera input.

2.3 Millimeter-Wave Radar And Multimodal Assistive Sensing

Millimeter-wave (mmWave) radar has been explored as an assistive sensing modality for visually impaired mobility because it can provide range and motion-related information without relying on visible light. Radar-based assistive systems can detect surrounding objects and support obstacle awareness in navigation tasks. For example, Kiuru et al. presented a wearable mmWave radar-based electronic travel aid and evaluated it with visually impaired participants, showing the potential of radar sensing for portable mobility assistance and user feedback [6]. Cardillo et al. further proposed a radar-cane system that integrates mmWave radar with a traditional white cane to warn users of potential obstacles and distinguish humans from non-human targets [2]. Recent studies have explored low-cost mmWave radar for body-centric sensing and continuous human-motion monitoring in mobile and wearable applications [4].

Compared with camera-only systems, mmWave radar is less affected by illumination changes and can provide useful range and motion information.

Prior work has also investigated the fusion of radar and vision sensors for visually impaired navigation. Long et al. integrated a low-power mmWave radar with an RGB-D sensor, where radar provided position and velocity information and the vision module supported object recognition [8]. Their results showed that sensor fusion provided additional target-state information, extended the effective detection range, and increased stability under varying illumination conditions [8]. However, radar signals do not directly represent the semantic state of a pedestrian traffic light. For example, radar may indicate that an object is approaching, but it cannot determine whether the current pedestrian signal is red or green. This limitation motivates the combination of radar sensing and mobile vision in the proposed system.

Existing studies have addressed intersection accessibility, smartphone-based signal recognition, crosswalk detection, radar-based obstacle sensing, body-centric radar sensing, and radar–vision fusion. Few prior studies combine distance-aware obstacle perception, lightweight pedestrian-signal recognition, edge processing, and smartphone-based feedback specifically for intersection crossing assistance. GuideSense addresses this gap by integrating complementary radar and vision sensing in a unified prototype for intersection crossing assistance.

3 Materials And Methods

GuideSense is an assistive navigation system for blind and visually impaired users that integrates an external millimeter-wave (mmWave) radar with a smartphone camera to provide obstacle awareness and pedestrian-signal recognition. Inspired by the conventional three-layer IoT architecture [9], GuideSense is organized into a perception layer, a network and edge processing layer, and an application layer. The mmWave radar and smartphone camera collect environmental information, while a Raspberry P_i performs radar analysis, pedestrian-signal recognition, and rule-based multimodal decision-making. The resulting safety information is then transmitted to the smartphone, which provides vibration and voice feedback to the user.

3.1 Perception Layer

The mmWave radar and smartphone camera are the two primary sensing components used by GuideSense to perceive the surrounding environment. The radar provides obstacle distance and motion information, while the smartphone camera provides visual semantic information about pedestrian signals.

3.1.1 mmWave Radar

This study employs the K60168A Dongle mmWave radar module developed by KaiKuTeK [7]. The processed radar information includes distance and relative velocity. Distance information is used to determine the proximity of detected obstacles, while relative velocity provides information about their motion with respect to the user. The radar measurements are represented as distance–velocity feature maps, in which the vertical axis represents target distance and the horizontal axis represents relative velocity. The intensity of each region indicates the strength of the radar response. An example distance–velocity feature map is shown in Figure 1. The system uses these features to determine the distance and motion characteristics of obstacles ahead.

3.1.2 Smartphone Camera

The mmWave radar mainly provides obstacle distance and motion information but cannot interpret the semantic meaning of pedestrian signals. Therefore, GuideSense uses the smartphone's built-in camera to provide visual information and assist blind and visually impaired users in identifying pedestrian-signal states when approaching or waiting at intersections. The smartphone camera continuously captures images of the environment ahead. These images are subsequently transmitted to the Raspberry P_i for pedestrian-signal recognition. The system focuses on recognizing red and green pedestrian signals, which provide important visual information for subsequent crossing assistance and safety decision-making [3].

3.2 Network And Edge Processing Layer

The network and edge processing layer is responsible for transmitting sensing data from the mmWave radar and smartphone camera to the Raspberry P_i and processing the information from both sensing modalities. As shown in Figure 2, the mmWave radar transmits obstacle-sensing data to the Raspberry P_i through a USB interface, while the smartphone camera sends pedestrian-signal images over Wi-Fi using HTTP. For radar processing, the received

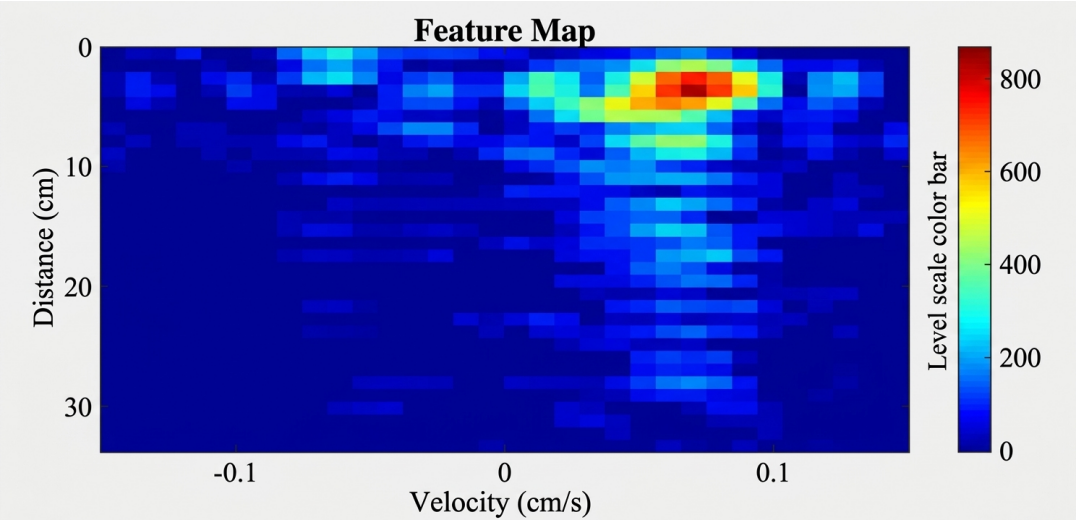


Figure 1: Example distance–velocity feature map provided with the K60168A radar development materials. Reproduced from [7]. The color intensity represents the radar response strength at different target distances and relative velocities.

distance–velocity feature maps are used as inputs to a lightweight 3D CNN for distance classification. Radar feature maps collected at different obstacle positions are used to train the model. During operation, the model assigns each incoming feature map to one of three distance states and returns the associated confidence score.

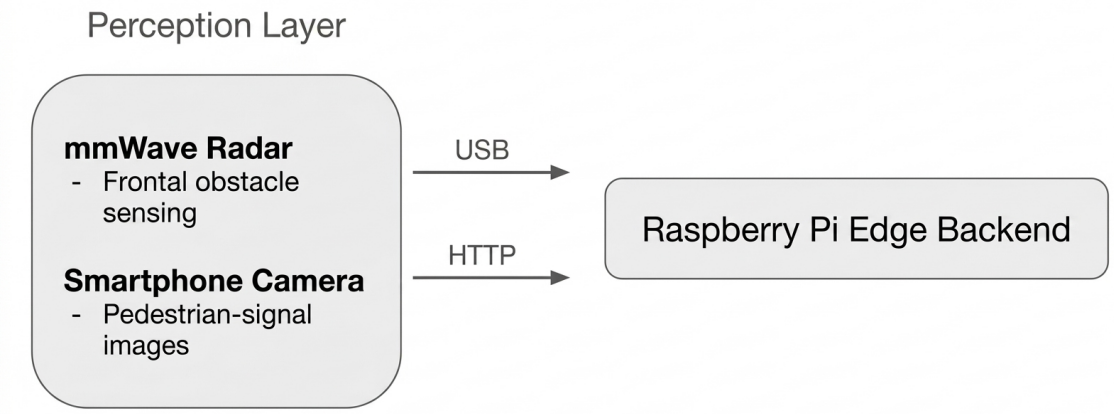


Figure 2: Data transmission from the sensing layer to the Raspberry P_i edge backend. The mmWave radar transmits obstacle-sensing data through USB, while the smartphone camera sends pedestrian-signal images over Wi-Fi using HTTP.

The radar output is represented using three distance states: Close, Far, and Background. The Close state covers obstacles at distances of $0 \leq d < 100$ cm, whereas the Far state covers obstacles at distances of $100 \leq d \leq 240$ cm. The third state is Background. The Background state represents either the absence of a detected approaching obstacle or an obstacle located beyond 240 cm. Therefore, a Background state should not be interpreted as a guarantee that the entire walking path is free of hazards.

Meanwhile, the smartphone camera continuously captures images of the environment ahead and transmits them to the Raspberry P_i over Wi-Fi using the HTTP communication protocol. Upon receipt, the images are processed on the Raspberry P_i using YOLOv8n. The model identifies the pedestrian signal as red or green and reports the corresponding confidence score, allowing the system to distinguish between red and green pedestrian signals [5, 11].

A rule-based multimodal decision module then combines this result with the radar-derived obstacle distance state to determine the current safety state. Only the safety decision is transmitted to the mobile application for user feedback; raw radar signals and image-processing data remain on the Raspberry Pi.

3.3 Application Layer

As shown in Figure 3, GuideSense integrates the perception, network and edge processing, and application layers into a complete processing pipeline. The application layer delivers the safety decision from the Raspberry P_i to the user through smartphone-based feedback. After rule-based multimodal decision-making is completed on the Raspberry Pi, the resulting safety state is transmitted to the GuideSense mobile application. The received safety state is mapped to different feedback cues. When a green pedestrian signal is recognized and no nearby obstacle posing an immediate collision risk is detected, the system provides a crossing assistance cue. When a red pedestrian signal is recognized or a nearby obstacle is detected, the system provides a stop or danger warning.

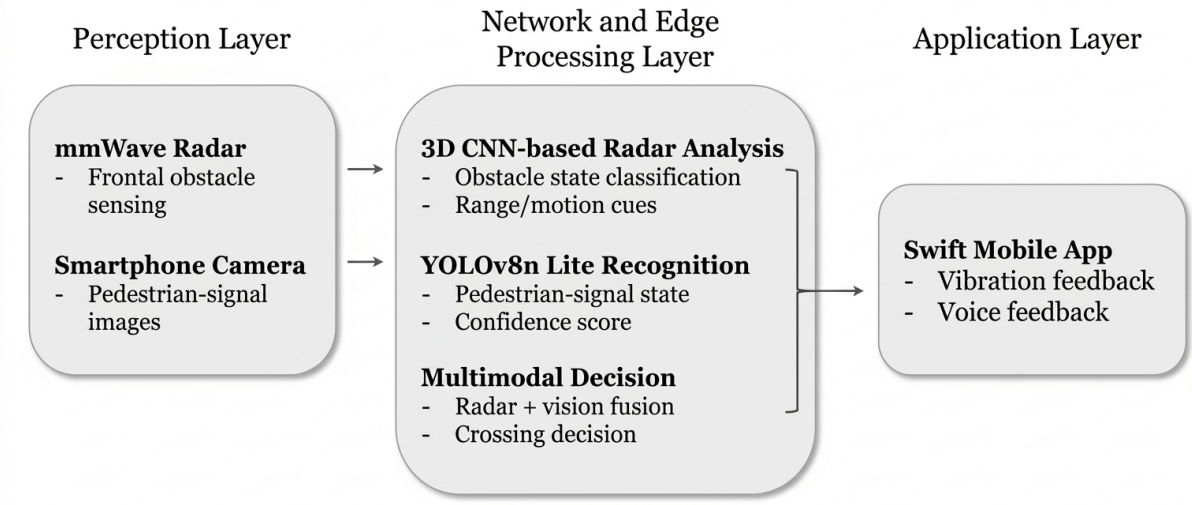


Figure 3: System overview of GuideSense. The mmWave radar provides frontal obstacle sensing, while the smartphone camera captures pedestrian-signal images for YOLOv8n recognition. The Raspberry Pi-based edge backend performs radar analysis, pedestrian-signal recognition, and multimodal decision-making. The resulting safety decision is then transmitted to the Swift mobile application, which provides vibration and voice feedback for intersection crossing assistance.

Auditory and tactile cues were selected as complementary nonvisual feedback modalities, motivated by prior research on audio-tactile integration [10]. Voice prompts provide information about the current pedestrian-signal state, such as whether the detected signal indicates a crossing or stopping condition. Vibration feedback is used to indicate obstacle-related warnings. In the current prototype, closer obstacles are represented by more frequent vibration feedback, while farther obstacles are represented by less frequent vibration feedback.

In this design, the smartphone serves as the final user-interface component rather than the main processing unit. The computationally intensive radar analysis and pedestrian-signal recognition are performed on the Raspberry Pi, while the smartphone focuses on image capture, feedback delivery, and user interaction. GuideSense converts the processed radar and visual information into auditory and tactile cues for intersection crossing assistance.

4 Experiment Results

4.1 Experimental Setup And Evaluation Protocol

This section evaluates the implemented GuideSense prototype through pedestrian-signal recognition, radar-based distance classification, and integrated system testing. The hardware configuration consists of an mmWave radar module, a Raspberry Pi-based edge computing device for radar processing and YOLOv8n pedestrian-signal recognition, and a smartphone for image capture and user feedback. The visual recognition model, YOLOv8n, was trained using a custom dataset of 1,000 annotated pedestrian-signal images, including 800 training images and 200 validation images across two target classes. The radar subsystem uses a lightweight 3D CNN to classify radar observations into three distance states: Close ($0 \leq d < 100$ cm), Far ($100 \leq d \leq 240$ cm), and Background (either the absence of a detected approaching obstacle or an obstacle located beyond 240 cm). Recurrent layers such as LSTM are excluded to reduce computational overhead for edge deployment. The edge device transmits the radar state to the smartphone through HTTP requests at an update interval of approximately 0.2 seconds. The integrated prototype was evaluated through blindfolded simulation testing, including five indoor trials and five outdoor trials in intersection-like

environments. These tests evaluated the interaction among sensing, communication, and user feedback; they did not evaluate effectiveness for visually impaired users.

4.2 Experimental Results

The results cover pedestrian-signal recognition, radar-based distance classification, and integrated prototype evaluation. The first two parts evaluate the individual sensing modules, while the integrated evaluation examines whether perception, edge communication, and smartphone feedback can operate together in blindfolded simulation scenarios.

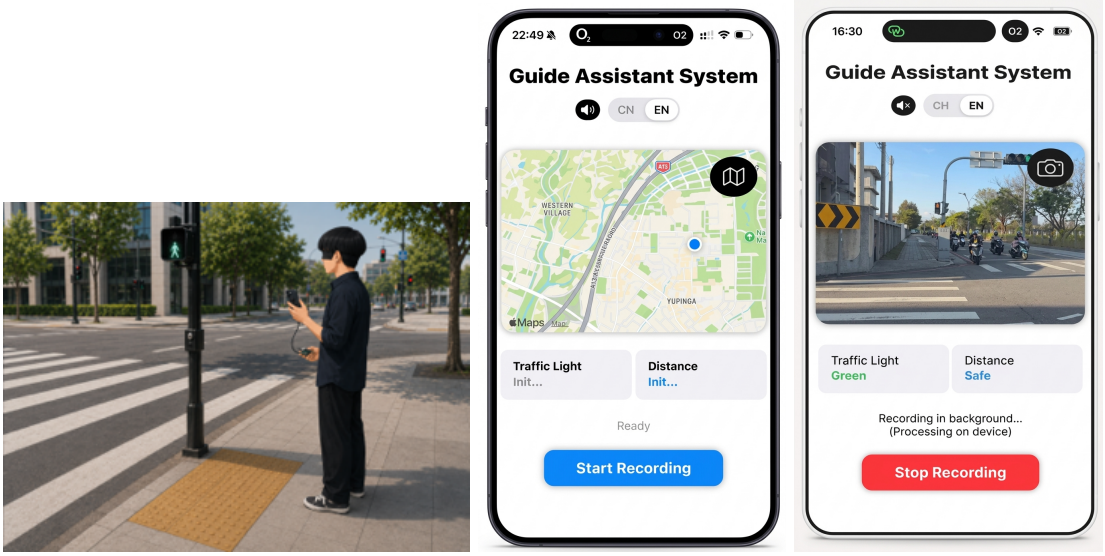
4.2.1 YOLOv8n Results

As summarized in Table 1, the YOLOv8n model achieved an mAP@0.5 of 98.1% on the 200-image validation set, indicating that the lightweight detector can recognize the target pedestrian-signal classes under the evaluated dataset conditions. Based on the validation performance, a selected confidence threshold of approximately 0.31 was used to reduce low-confidence detections caused by background objects, glare, or partial occlusion while retaining sensitivity to target signals. The model performed well within the experimental dataset, but its robustness requires further evaluation under more diverse lighting, weather, camera-angle, and real-world intersection conditions.

4.2.2 Radar Classification Results

As summarized in Table 2, the radar subsystem was evaluated using three distance states: Close, Far, and Background. The lightweight 3D CNN achieved an overall precision of 90.5%, recall of 89.9%, and macro F1-score of 89.8% across the evaluated samples. The Close state corresponds to obstacles within 0–100 cm and is treated as a high-priority warning state because nearby obstacles require immediate feedback. The Background state indicates that no approaching obstacle was detected within the defined warning range. However, it should not be interpreted as a guarantee that the entire walking path is free of hazards.

The results show that the radar module provides obstacle-state information for the GuideSense prototype. Radar-based classification complements the visual recognition module by providing obstacle-distance information that pedestrian-signal recognition alone cannot provide. Boundary cases between Close and Far remain difficult to classify and require further refinement of radar data collection, model training, and distance-threshold design.



(a) Conceptual illustration of outdoor crossing trial. (b) UI design and real-time navigation module. (c) Outdoor testing environment.

Figure 4: Experimental environments and prototype deployment.

4.2.3 Integrated System Results

The integrated evaluation examined whether visual recognition, radar-based obstacle awareness, edge communication, and smartphone feedback could operate together during the five indoor and five outdoor blindfolded simulation trials.

With an HTTP update interval of approximately 0.2 seconds, the prototype generated vibration and voice feedback based on the detected pedestrian-signal state and radar obstacle state.

Table 1: YOLOv8n detection performance on the validation set

Metric	Result
Training images	800
Validation images	200
Number of classes	2
mAP@0.5	98.1%
Confidence threshold	0.31
Precision	99.3%
Recall	96.1%
F1-score	97.6%
Inference time	32ms

Table 2: Radar distance-classification performance

Class	Distance Range (cm)	Precision	Recall	F1-score
Close	$0 \leq d < 100$	92.0%	76.7%	83.6%
Far	$100 \leq d \leq 240$	79.4%	93.1%	85.7%
Background	> 240 or None	100%	100%	100%
Overall	–	90.5%	89.9%	89.8%

The results show that the two sensing modalities provide complementary information. Vision provides semantic information about pedestrian-signal states, while radar provides distance-related obstacle awareness. The prototype uses crossing-signal status and nearby obstacle conditions to generate user feedback. The controlled blindfolded simulation trials provide preliminary evidence that the system components can operate together.

4.3 Discussion And Error Analysis

To identify system limitations, we analyzed sensor failure cases and feedback behaviors observed during the blindfolded simulation tests. The rule-based multimodal decision strategy allowed the prototype to combine pedestrian-signal recognition and radar-based obstacle awareness when generating user feedback. In particular, the system was designed to avoid issuing a positive crossing cue when the pedestrian signal was not confidently recognized or when a close obstacle was detected. The conservative feedback design prevents a "Go" instruction when the sensing evidence is incomplete or conflicting, as required for safety-critical assistive scenarios.

Several limitations must be acknowledged. First, the visual model was trained on a limited dataset of 1,000 annotated pedestrian-signal images, which may not fully cover diverse lighting, weather, camera angles, and real-world intersection conditions. Second, the integrated evaluation relied on sighted participants undergoing blindfolded simulation rather than actual visually impaired users. Third, the current prototype evaluation focused on technical feasibility and module interaction, rather than clinical effectiveness or real-world safety.

The current results provide preliminary evidence that the GuideSense prototype can integrate sensing, communication, and feedback at an update interval of approximately 0.2 seconds. However, they do not yet demonstrate clinical effectiveness or real-world safety for visually impaired users. Future work should expand the visual and radar datasets and evaluate the system in more diverse intersection environments. It should also refine the multimodal decision strategy and conduct IRB-approved studies with visually impaired participants under controlled and supervised conditions.

Practical Implications. The results suggest that combining lightweight vision with mmWave radar can provide complementary semantic and distance information while keeping computationally intensive processing on an edge device. This architecture may serve as a practical basis for future portable assistive navigation systems, although real-world safety must still be validated with visually impaired users.

5 Conclusions

This paper presented GuideSense, an implemented multimodal assistive navigation prototype for intersection crossing assistance for pedestrians with visual impairments. The system integrates mmWave radar-based obstacle perception,

YOLOv8n pedestrian-signal recognition, Raspberry Pi-based edge processing, and smartphone-based vibration and voice feedback. GuideSense integrates visual semantic information for pedestrian-signal understanding with radar-based distance information for nearby obstacle awareness.

Under controlled testing conditions, the experimental results show that the implemented system components can operate together. YOLOv8n achieved an mAP@0.5 of 98.1% on the validation set, and the radar-based 3D CNN achieved an overall precision of 90.5%, recall of 89.9%, and macro F1-score of 89.8%. The indoor and outdoor blindfolded simulation trials showed that the sensing modules, edge communication, and smartphone feedback operated together in the prototype.

Although the prototype has been implemented and evaluated, the current study remains limited to controlled simulation testing with sighted participants. Future work will expand the visual and radar datasets and evaluate the system in more diverse intersection environments. It will also refine the multimodal decision strategy and conduct supervised studies with visually impaired participants.

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Biography

Ching-Min Huang is a senior undergraduate student in the Department of Information Engineering and Computer Science at Feng Chia University, Taiwan. Her research interests include computer vision, sensor fusion, and AI-based assistive technologies, with a particular interest in developing navigation systems for people with visual impairments.

Shih-Ming Wang is a senior undergraduate student in the Department of Information Engineering and Computer Science at Feng Chia University, Taiwan. His research interests include computer vision, machine learning, large language models, intelligent detection systems, and artificial intelligence applications.

Min-Yen Kuo is a senior undergraduate student in the Department of Information Engineering and Computer Science at Feng Chia University, Taiwan. Her research interests include artificial intelligence, intelligent sensing,

millimeter-wave radar applications, assistive technology, and the Internet of Things.

Chun-Hsiu Yeh received a Ph.D. degree in Information Engineering from National Chung Hsing University in 2017 and an M.S. degree in Information Management from Chaoyang University of Technology in 2001. She is currently an associate professor in the Department of Information Engineering and Computer Science at Feng Chia University. Her current research interests include image processing, computation of information systems, and intelligent detection issues.

Xian-Zhong Lin received an M.S. degree in Information Management from Chaoyang University of Technology in 2007. He is currently pursuing a Ph.D. degree in Information Engineering and Computer Science at Feng Chia University. His research interests include image processing, deep learning, and intelligent detection.